CONFORMAL DIMENSION OF EXPONENTIAL METRICS IN $d \geq 2$

ABSTRACT. Let $d \geq 2$ and $0 < \gamma < \sqrt{2d}$. We prove that the ordinary conformal dimension of the subcritical γ-exponential metric on $\mathbb{R}^d$ is d. For $d \geq 3$ we use the higher-dimensional metric of [1], while for $d = 2$ we use the usual Liouville quantum gravity metric. The proof uses the good-annulus and hyperbolic-filling method of Miller and Tian, with a Brownian time-change argument for the required multiscale estimates. A circle-average process in a fixed two-dimensional plane gives the exact exponent in every ambient dimension. We also deduce that the conformal dimension of the γ-Liouville quantum gravity sphere is two for all $\gamma \in (0, 2)$.

This extension is a research argument and has not been independently refereed.

1. INTRODUCTION

For a metric space (X, D), its *ordinary conformal dimension* is

$$\mathrm{Cdim}(X, D) = \inf\{\dim_{\mathrm{H}}(X, \delta) : \mathrm{id} : (X, D) \to (X, \delta) \text{ is quasisymmetric}\}. \tag{1}$$

Let h be the whole-space log-correlated Gaussian field on $\mathbb{R}^d$, defined modulo constants by the covariance kernel $-\log|x - y|$. For $d = 2$ this is the whole-plane Gaussian free field. For $d \geq 3$ it is not the ordinary d-dimensional Gaussian free field. Fix $0 < \gamma < \sqrt{2d}$ and write

$$Q = \frac{d}{\gamma} + \frac{\gamma}{2}, \qquad \xi = \frac{\gamma}{d_\gamma}, \qquad a = Q - \gamma = \frac{d}{\gamma} - \frac{\gamma}{2} > 0, \tag{2}$$

where d_γ is the Hausdorff dimension of the associated exponential metric. We let D_h^γ denote the strong subcritical exponential metric of [1, Definition 1.1 and Theorem 1.2] for $d \geq 3$ and the canonical γ-LQG metric of [5] for $d = 2$. Our arguments require only the metric properties and estimates stated in Section 2. The strong higher-dimensional characterization is itself presently available as a research manuscript; the present result should be read with that input understood.

Theorem 1.1. *For every integer $d \geq 2$ and every $\gamma \in (0, \sqrt{2d})$,*

$$\mathrm{Cdim}(\mathbb{R}^d, D_h^\gamma) = d \qquad \textit{almost surely.} \tag{3}$$

More precisely, for each fixed $p > d$, almost surely there is a metric δ_p on all of $\mathbb{R}^d$ such that $\mathrm{id} : (\mathbb{R}^d, D_h^\gamma) \to (\mathbb{R}^d, \delta_p)$ is quasisymmetric and $\mathcal{H}^p_{\delta_p}(\mathbb{R}^d) = 0$.

The statement concerns each fixed γ. It does not assert that the infimum in the definition of conformal dimension is attained or that the exponential metric is quasisymmetric to the Euclidean metric.

The proof follows the good-annulus and hyperbolic-filling strategy of Miller and Tian [2, Sections 5–6], who used it to determine the conformal dimension of the Brownian sphere. We give the deterministic argument in full and do not use their conformal-dimension theorem. The probabilistic input is different: it comes from an exact spectral calculation for a stationary Brownian time change. Indeed, with

$$\lambda(p) = \sqrt{a^2 + 2p} - a, \tag{4}$$

the identity $Q^2 - a^2 = 2d$ gives

$$\lambda(d) = \gamma, \qquad \lambda(p)/\xi > d_\gamma \quad (p > d).$$

The strict inequality allows the losses from exceptional annular scales. We first obtain a rooted product estimate, then use Gaussian multiplicative chaos to integrate it over the field, and finally turn it into a Hausdorff covering bound for a single globally quasisymmetric metric.

The higher-dimensional geometric input is the distance between arbitrary crossings of a Euclidean annulus, which replaces the separating loops available in the plane. For filled metric balls we work with an intrinsic hull-arrival coordinate, since a filled ball may contain points at root-distance greater than its radius.

Throughout, $B_r^{\mathrm{E}}(x)$ denotes a Euclidean ball and $B_r^D(x)$ a ball for the metric under discussion. We write $D(E, F; U)$ for internal distance within U, and μ_h for the γ-Gaussian multiplicative chaos measure. Constant shifts satisfy $D_{h+c} = e^{\xi c} D_h$ and $\mu_{h+c} = e^{\gamma c}\mu_h$.

2. The exponential metric and its basic estimates

We use the metric $D_h = D_h^\gamma$ of [1] for $d \geq 3$ and [5] for $d = 2$. It is a measurable local length metric inducing the Euclidean topology; its internal metrics are determined locally by the field. It satisfies simultaneous Weyl scaling by continuous functions and affine covariance. In particular, adding a constant c multiplies distances by $e^{\xi c}$.

The normalized internal distance between fixed disjoint nontrivial continua in a fixed Euclidean domain has positive and negative moments of every order [3, Proposition 4.7]. The same is true of the *around distance*

$$U_D(A) = \sup_{P_1, P_2} D(P_1, P_2; A), \tag{5}$$

where the supremum ranges over paths crossing a fixed Euclidean annulus A [3, Proposition 4.10]. These estimates apply to the metric D_h^γ by its characterization and normalization in [1]; for $d = 2$ they are the standard LQG annular estimates. We never use all positive moments of an arbitrary metric diameter: the available diameter moment range is $b < 2dd_\gamma/\gamma^2$ [3, Proposition 4.11].

Let $h_r^{\mathrm{sph}}(z)$ be the spherical average of h. For a fixed shell shape A, fixed disjoint connected endpoint sets $K_1, K_2 \Subset U$, and every $b > 0$,

$$\sup_{z,r} \mathbb{E}\left[\left(r^{-\xi Q} e^{-\xi h_r^{\mathrm{sph}}(z)} U_D(z + rA)\right)^b + \left(r^{-\xi Q} e^{-\xi h_r^{\mathrm{sph}}(z)} D(z + rK_1, z + rK_2; z + rU)\right)^{\pm b}\right] < \infty. \tag{6}$$

As usual, a $\pm b$-power denotes the sum of the b- and $(-b)$-powers. Whenever an endpoint lies on the boundary of a shell, we work first in a slightly enlarged fixed shell and then trim the relevant crossing.

The metric covariance theorem in [1] is stated for each fixed affine map. Since the radial normalization below selects a dilation from the field, we use the following version simultaneously under all dilations.

Proposition 2.1 (An equivariant metric version)**.** *The metric D_h admits a version which agrees with it almost surely and for which*

$$D_h(sx, sy) = D_{h(s\cdot) + Q\log s}(x, y), \qquad s > 0, \tag{7}$$

holds along the full dilation orbit of a single field realization. Its local metric laws, Weyl scaling, and the moment estimates above are unchanged.

Proof. For $r > 0$ let $D_h^{[r]}(x, y) = r^{-\xi Q} D_{h(\cdot/r)}(rx, ry)$. Fixed-map affine covariance and constant Weyl scaling imply that $D_h^{[r]} = D_h$ almost surely for each deterministic r. By Fubini, the essential supremum

$$q_h(x, y) = \operatorname*{ess\,sup}_{r>0} D_h^{[r]}(x, y),$$

taken with respect to $\mathrm{d}r/r$, equals $D_h(x,y)$ almost surely, simultaneously on a countable dense set of pairs and hence, by continuity, on all pairs. Choose this continuous metric version. For every $s > 0$, the change of variables $r \mapsto sr$ and the constant-shift rule give directly

$$q_{h(s\cdot)+Q\log s}(x,y) = q_h(sx, sy).$$

This is an algebraic identity of the essential-supremum definition, so it holds for every s on the same dilation orbit, including a field-dependent choice of s. The metric remains local and satisfies the stated moment estimates since it agrees with the original metric almost surely. Simultaneous Weyl scaling is inherited by applying the same construction on the continuous-perturbation orbit, using the local comparison of Weyl-weighted lengths on compact paths. The same version can be used on a field with a logarithmic insertion: apply the construction on punctured subdomains by locality and take the unique metric completion at the insertion. The required existence of the completion follows from the summable inward annular upper bounds established below. □

3. A Brownian coordinate for the log-correlated field

Averaging over $(d-1)$-spheres does not give the exact Brownian process needed below. Instead, fix an oriented two-dimensional plane $E \subset \mathbb{R}^d$, and let σ_r^z be uniform probability measure on the circle $z + rS^1 \subset z + E$. Circle averages $h_r^\circ(z) = (h, \sigma_r^z)$ are well-defined by finite logarithmic energy; this does not require restricting h to a sample path or to an arbitrary codimension-one hypersurface.

Lemma 3.1 (Brownian circle averages)**.** *Normalize the field by $h_1^\circ(0) = 0$. Then $B_t = h_{e^t}^\circ(0)$, $t \in \mathbb{R}$, is a continuous two-sided standard Brownian motion started at zero. Moreover,*

$$\theta_t h = h(e^t \cdot) - h_{e^t}^\circ(0)$$

form a measure-preserving flow on the normalized-field law. Local measurable functionals of this flow are stationary in physical logarithmic time.

Proof. For circles in the same two-dimensional plane, logarithmic potential theory gives

$$\iint \log|x-y|\, \sigma_{e^s}^0(\mathrm{d}x)\sigma_{e^t}^0(\mathrm{d}y) = \max(s,t).$$

Apply the covariance of the field to the signed, total-mass-zero measures $\sigma_{e^s}^0 - \sigma_1^0$ and $\sigma_{e^t}^0 - \sigma_1^0$. This gives

$$\mathbb{E}[B_s B_t] = -\max(s,t) + \max(s,0) + \max(t,0) = \frac{|s| + |t| - |s-t|}{2}.$$

This is precisely the covariance of two-sided Brownian motion. Gaussian continuity gives the continuous version. Covariance invariance under translation in logarithmic scale, modulo field constants, proves invariance under θ_t and the flow identity $\theta_s\theta_t = \theta_{s+t}$. The remaining field need not be independent of B. □

Changing the spherical average to this circle average does not alter any moment range. Indeed $h_r^{\mathrm{sph}}(z) - h_r^\circ(z)$ has a centered Gaussian law of bounded variance independent of (z,r), by scaling of the covariance kernel. Consequently it has exponential moments of every order. Hölder's inequality converts (6) to the normalization $r^{\xi Q} e^{\xi h_r^\circ(z)}$. The same observation applies to normalized GMC masses and the finite moment range for diameters.

Lemma 3.2 (Annular Gaussian product inequality)**.** *For $b, H > 0$, set $A_b = \{e^{-b} < |x| < e^b\}$. There is $p_0 = p_0(d,b,H) < \infty$ such that if $F_k \geq 0$ is measurable with respect to $(\theta_{kH}h)|_{A_b}$, then for each finite $I \subset \mathbb{Z}$,*

$$\mathbb{E}\prod_{k\in I} F_k \leq \prod_{k\in I} (\mathbb{E}F_k^{p_0})^{1/p_0}. \tag{8}$$

The statement is unchanged by deterministic local mean shifts. If the windows are spaced by $L \to \infty$, their common exponent can be chosen to tend to one.

Proof. Each centered Gaussian functional of the restriction has a representation by a signed measure of total mass zero, supported in a fixed compact annulus. Such a measure has covariance norm given, up to a dimensional factor, by

$$\|\nu\|_h^2 = \int_{\mathbb{R}^d} |\widehat{\nu}(\zeta)|^2 |\zeta|^{-d} \, \mathrm{d}\zeta.$$

For two windows whose logarithmic centers are separated by $T > 2b + 2$, rescale so their supports have relative scale $\rho = e^{-T}$. Using the zero-mass condition, their covariance is a bilinear form with kernel

$$K_\rho(x, y) = \chi(x)\chi(y)\{-\log|x - \rho y| + \log|x|\},$$

where χ is a smooth cutoff equal to one on the relevant compact supports. Every derivative of K_ρ is $O(\rho)$. Fourier duality between the covariance Sobolev space and its dual bounds the maximal Gaussian correlation of the two window spaces by Ce^{-T}. For overlapping windows use the bound one. The resulting correlations ρ_j satisfy $\sum_{j \geq 1} \rho_j < \infty$.

For every finite collection of block Gaussian variables, Cauchy–Schwarz and $2uv \leq u^2 + v^2$ give the covariance matrix comparison

$$\Sigma \leq p_0 \, \mathrm{diag}(\Sigma_{kk}), \qquad p_0 > 1 + 2 \sum_{j \geq 1} \rho_j.$$

Apply the Gaussian Hölder inequality of [7, Theorem 1(i)]. Finite-dimensional approximations and monotone convergence give (8) for arbitrary nonnegative block functionals. Deterministic shifts affect means, not the covariance matrix. For L-separated windows the correlation row sum is $O(e^{-L})$. □

Lemma 3.3 (Uniform circle averages and exponential integrals)**.** *For a smooth additive normalization and every compact $K \subset \mathbb{R}^d$ and $\varepsilon > 0$, almost surely*

$$\sup_{x \in K} |h_r^\circ(x)| \leq (\sqrt{2d} + \varepsilon) \log r^{-1} \quad \text{for all sufficiently small } r. \tag{9}$$

The circle averages admit a jointly continuous version on $\mathbb{R}^d \times (0, \infty)$. If $Q > \sqrt{2d}$, then

$$J_x(t) = \int_{-\infty}^{t} e^{\xi h_{e^s}^\circ(x) + \xi Q s} \, \mathrm{d}s \tag{10}$$

is finite, positive, jointly continuous in (x, t), and tends to zero uniformly as $t \to -\infty$ for x in compact sets. The conclusions persist under continuous field perturbations.

Proof. The circle measure has finite β-Riesz energy for $0 < \beta < 1$. The Fourier covariance representation implies $\mathbb{E}[(h_1^\circ(x) - h_1^\circ(y))^2] \leq C_\beta |x - y|^\beta$; scaling gives the analogous estimate at radius r. In logarithmic radius the increment variance is $|\log(r/s)|$. Gaussian moment bounds and Kolmogorov's criterion give joint continuity away from $r = 0$.

At $r = 2^{-m}$, a grid of mesh r contains $O(r^{-d})$ points in a fixed compact set. Their average variances are $\log r^{-1} + O(1)$. Gaussian tails show that the probability any grid value exceeds $b \log r^{-1}$ is summable over m whenever $b > \sqrt{2d}$. The oscillations of $h_{rv}^\circ(z + ru) - h_r^\circ(z)$ for (u, v) in a fixed compact set have a uniformly bounded Gaussian supremum with Gaussian tails, by the increment estimate and Borell's inequality. A union bound over the same grid controls interpolation errors of order $\varepsilon \log r^{-1}$; these probabilities are summable. This proves (9) for all small radii.

Choose ε so that $Q > \sqrt{2d} + \varepsilon$. For sufficiently negative s, the integrand in (10) is at most $e^{\xi(Q - \sqrt{2d} - \varepsilon)s}$ uniformly over compact sets of centers. Dominated convergence proves all remaining

assertions. A continuous perturbation has uniformly bounded circle averages over these sets and radii. □

4. Filled balls and geometric shell estimates

This section is deterministic except when shell moments are invoked. Let D be a proper geodesic metric inducing the Euclidean topology. Properness for our metric follows from fixed-shell lower estimates and Borel–Cantelli, and a continuous proper length metric is geodesic. The same arguments apply to a root of logarithmic strength $\gamma < Q$ after completing the punctured space by one point.

Denote by $H_r^\circ(x)$ the complement of the unbounded component of $\mathbb{R}^d \setminus B_r^D(x)$, and by $H_r^\bullet(x)$ the complement of the unbounded component of $\mathbb{R}^d \setminus \overline{B_r^D(x)}$. The two conventions need not agree at an exceptional radius. The former is open, the latter compact, and both are bounded and connected. Filling is always towards infinity.

Lemma 4.1 (Filled-ball topology)**.** *The boundaries $\partial H_r^\circ(x)$ and $\partial H_r^\bullet(x)$ are connected compact separators of x from infinity. Every point of either boundary has D-distance exactly r from x. For $0 < r < R$, the mixed band*

$$A_{r,R}(x) = H_R^\circ(x) \setminus H_r^\bullet(x) \tag{11}$$

is open and connected, has two connected complementary components in the one-point compactification, and its internal distance from its inner boundary to each point of its outer boundary is $R - r$.

Proof. Metric balls are path connected, and filling preserves connectedness. Sonneborn's connected-boundary lemma [8, Lemma 1] in S^d, $d \geq 2$, applied to the exterior complementary component gives boundary connectedness. A boundary point of a filled ball cannot have distance below r, since it would lie in the interior of the ball. If its distance were above r, a small connected neighborhood avoiding the closed ball would lie wholly in one complementary component, contradicting that it is a boundary point. This proves the boundary-level identity.

The inner and outer complementary continua of (11) are disjoint. Each is nonseparating, and Alexander duality (or the elementary connected-complement argument on S^d) gives connectedness of the complement of their union. It is open, hence path connected. An inward geodesic from the outer boundary crosses the levels r and R in order. Its terminal segment of length $R - r$ remains inside the mixed band: after crossing level r it cannot reenter and leave the filled inner hull, since any subsequent exit would have root distance r; the corresponding observation holds at the outer hull. The triangle inequality gives the reverse inequality. Boundary completion points follow by interior approximation. □

4.1. Filled annuli and the intrinsic arrival coordinate.

For $t, s > 0$ put

$$A = A(x; t, s) := H_{t+8s}^\circ(x) \setminus H_t^\bullet(x), \qquad I = \partial H_t^\bullet(x), \quad O = \partial H_{t+8s}^\circ(x).$$

It is important not to identify the middle of A with a difference of filled balls of intermediate radii. A filled ball can contain points whose distance from the center exceeds its radius, because it also contains bounded complementary components. We instead use the intrinsic metric of A.

For $z \in A$ write $d_A(I, z)$ for the infimum of lengths of paths in A joining z to points approaching I. For $0 < q < 8s$ let N_q be the open d_A-neighborhood of I. Adjoin the inner complementary continuum to N_q and fill all components of its complement which are disjoint from the outer complementary continuum. Denote the resulting open, inner-directed filled set by $\mathfrak{H}_q$, using the closed convention $\mathfrak{H}_q^\bullet$ when necessary. Equivalently, $\mathfrak{H}_q$ is the complement, within the sphere

compactification, of the component of the complement of N_q which meets O; all definitions can be taken with strict radius margins. Put

$$T_A(z) = \inf\{q : z \in \mathfrak{H}_q\}, \qquad z \in A.$$

This is the *intrinsic hull-arrival coordinate*. It extends continuously to I and O with boundary values 0 and $8s$, respectively.

Lemma 4.2 (Arrival levels and short balls)**.** *The coordinate T_A is 1-Lipschitz with respect to the internal metric of A. For $0 < u < v < 8s$, every path from $\mathfrak{H}_u^\bullet$ to $A \setminus \mathfrak{H}_v$ has length at least $v - u$. If $T_A(z) \in [2s, 6s]$, every ambient D-path of length less than s starting at z stays in A. In particular, the ambient and internal open s-balls about z agree and do not meet $I \cup O$.*

Proof. The argument is the filled-ball arrival argument, carried out in the inner metric of A. A point on the boundary of $\mathfrak{H}_q$ has internal distance q from I: smaller distance puts it in the interior of N_q, while a larger distance gives a connected neighborhood of the point in the complement of N_q, placing it entirely in one of its complementary components. Suppose $T_A(z) < u < v < T_A(w)$. Every path from z to w leaves $\mathfrak{H}_u$ at a point b with $d_A(I, b) = u$. Since w is outside $\mathfrak{H}_v$, $d_A(I, w) \geq v$. The remainder of the path has length at least $v - u$ by the triangle inequality for the inner distance. This proves both Lipschitz continuity and separation of the closed and open layers. By Lemma 4.1, the distance from I to each point of O in the internal completion is $8s$, so the claimed boundary values follow. A path from z to the complement of A has an initial segment in A terminating at I or O. Since $T_A(z) \in [2s, 6s]$, the Lipschitz bound makes that segment have length at least $2s$. This proves the last claim, even when z belongs to a component swallowed by some filled layer. □

Call the mixed band A *good* if, for every $z \in A$ with $T_A(z) \in [2s, 6s]$, the inner boundary I is in the component of $\mathbb{R}^d \setminus B_s^D(z)$ containing the outer complementary continuum (equivalently, containing infinity). Lemma 4.2 ensures that the ball in this definition is contained in A. The definition uses no assumptions on the topology of d_A-balls.

Lemma 4.3 (Failure of goodness forces a short separator)**.** *If $A(x; t, s)$ is not good, then* $\operatorname{diam}_D(O) \leq 18s$.

Proof. There is a z in the intrinsic middle layer such that the ambient s-ball about z separates I from O. By Lemma 4.2 it is contained in A. For any two points of O, their radial geodesics from x have terminal subarcs of length $8s$ within A joining I to O. Each such subarc meets the s-ball. Two meeting points can be connected through z with length less than $2s$. Concatenating at most the two terminal subarcs and this connection gives $\operatorname{diam}_D(O) \leq 8s + 2s + 8s = 18s$. □

Let $R_j = R_0\alpha^j$, $s_j = R_0\alpha^{j-1+\zeta}$, where $\alpha \in (0, 1)$ and $\zeta \in (0, 1)$. Define $F(x, j)$ to mean that $A(x; t, s_j)$ is good for some

$$R_{j-1}/8 \leq t < t + 8s_j \leq R_{j-1}/4.$$

Define $\widetilde{F}(x, j)$ to mean that *some* good band $A(o; t, s_j)$ satisfies

$$H^\circ_{R_{j-1}/16}(x) \subset H^\bullet_t(o), \qquad H^\circ_{t+8s_j}(o) \subset H^\circ_{5R_{j-1}/16}(x). \tag{12}$$

Only the following measurable sufficient condition is used in probabilistic integrals:

$$F_*(x, j) = \{\operatorname{Leb}\{v \in [3R_{j-1}/16, R_{j-1}/4] : \operatorname{diam}_D \partial H^\circ_v(x) > 32s_j\} > 0\}. \tag{13}$$

If $s_j \leq R_{j-1}/128$, Lemma 4.3 implies $F_* \subset F$, using $t = v - 8s_j$. The inclusion properties of filled balls give

$$F_*(x, j) \implies F(x, j) \implies \widetilde{F}(y, j) \quad \text{whenever } D(x, y) \leq R_{j-1}/16. \tag{14}$$

The same mixed band witnesses the last implication; strict radius margins give (12). Thus all probabilistic events remain the explicit Borel events F_*, and the extra intrinsic coordinate is only a deterministic ingredient.

4.2. **Shell connectors and integral envelopes.** Fix the Euclidean shell $A = A_{1/4,4}(0)$. A crossing of a smaller concentric shell has finite length by the set-to-set estimates. Two crossings of an overlapping shell can be joined within that overlap at cost at most U_D of (5). This is the replacement for the fact that certain planar crossings must intersect.

Let $A = A_{r,R}$ and let P be a path crossing its closed radial span, with interior in A and D-length at most L. Let $U = U_D(A)$, with boundary endpoints interpreted by interior approximation, and set

$$K = \overline{A} \cap \{z : D(z, P) \leq U + \varepsilon\}.$$

This set is compact by properness. Any path from the inner Euclidean ball to infinity has a subpath crossing the same shell A. The definition of U gives a pair of points, one on each crossing, at internal distance at most $U + \varepsilon/2$; hence that path meets K. Thus K separates the inner Euclidean ball from infinity. If P is reachable from the root at cost at most S, every point of K can be reached with cost at most $S + L + U + 2\varepsilon$. Filling the metric ball of this radius therefore contains the inner Euclidean ball. The shell moment estimate applies by choosing the paths and their endpoint sets in a fixed slightly larger domain, then trimming at the indicated radii. This argument uses no planar separating loop.

Choose a uniform phase $\Theta \in [0, 1)$, independent of the field, and set $a_k = k + \Theta$. At logarithmic Euclidean scale a_k, introduce a lower mark L_k equal to the distance across a fixed shell strictly inside $[a_k, a_k + 1]$. Introduce an upper mark C_k, a fixed finite sum of across and around distances in enlarged overlapping shells. Short crossings joined by the around marks give a path from the root to the outer crossing, and the preceding separator observation gives

$$\sum_{j \leq k-1} L_j \leq D(0, \partial B^{\mathrm{E}}_{e^{a_k}}(0)), \qquad B^{\mathrm{E}}_{e^{a_{k+1}}}(0) \subset H^{\circ}_{\sum_{j \leq k} C_j}(0). \tag{15}$$

At the rooted field $H = h - \gamma \log |\cdot|$, define

$$W_t = h^{\circ}_{e^t}(0) + at, \qquad Z(t) = e^{\xi W_t}. \tag{16}$$

Normalized marks $e^{-\xi W_{a_k}} L_k$ and $e^{-\xi W_{a_k}} C_k$ are measurable in fixed-width logarithmic windows and have all positive moments; the lower marks also have all negative moments. Since $W_t = at + o(|t|)$ as $t \to -\infty$, the inward sum of the upper marks is finite almost surely. The connecting path consequently converges to the root in the metric completion.

The placement of the discrete marks in the continuous envelopes matters. For each $k \in \mathbb{Z}$ and $t \in [a_k, a_{k+1})$, set

$$x(t) = C_{k+1} e^{-\xi W_t}, \qquad y(t) = L_{k-1} e^{-\xi W_t}.$$

Equivalently, $y(t) = L_k e^{-\xi W_t}$ on $[a_{k+1}, a_{k+2})$. These are fixed-width local normalized marks. The upper mark for shell $k + 1$ is recorded on the preceding interval, and the lower mark for shell $k - 1$ on the succeeding interval. Their integrals over each full interval give exactly the corresponding discrete shell marks. Define the envelopes

$$A(t) = \int_{-\infty}^{t} Z(s) y(s) \, \mathrm{d}s, \qquad S(t) = \int_{-\infty}^{t} Z(s) x(s) \, \mathrm{d}s \tag{17}$$

satisfy, for every real t,

$$A(t) \leq D(0, \partial B^{\mathrm{E}}_{e^t}(0)), \qquad B^{\mathrm{E}}_{e^t}(0) \subset H^{\circ}_{S(t)}(0). \tag{18}$$

Indeed, for $t \in [a_k, a_{k+1})$, the upper integral contains all C_j with $j \leq k$, which by (15) enclose $B^{\mathrm{E}}_{e^{a_{k+1}}} \supset B^{\mathrm{E}}_{e^t}$. The lower integral contains all L_j with $j \leq k - 2$ and at most one further fractional contribution from L_{k-1}. Each of these crossings is contained in $B^{\mathrm{E}}_{e^t}$, so the fractional contribution

cannot invalidate the lower bound. The enlarged shell supports and Gaussian oscillations on unit logarithmic intervals give, for each $b > 0$,

$$\sup_k \mathbb{E}\left[\sup_{t\in[k+\Theta,k+1+\Theta]} \max\{x(t), y(t)^{-1}, 1\}^b\right] < \infty. \tag{19}$$

The complete normalized environment, with translated uniform phase, is stationary under deterministic shifts of physical logarithmic time.

4.3. **A deterministic obstruction to thin hull boundaries.** Set $r_k = 2^{-k}$ for $k \geq 0$. As above there are paths Q_k in $\{r_k/4 < |z| < 4r_k\}$ whose concatenation runs from the root beyond the unit Euclidean sphere, and whose lengths are at most positive local marks U_k with all normalized positive moments. There are also positive local marks ℓ_k with all normalized negative moments such that

$$\operatorname{diam}_D(K) \geq \ell_k \tag{20}$$

for every compact separator K of 0 and infinity meeting $\{r_k/4 < |z| < 4r_k\}$. Indeed, cover the latter shell by finitely many small Euclidean balls. If z lies on K, separation implies $0 \in \operatorname{conv}(K)$; hence some $w \in K$ has $|w - z| \geq |z|$. The pair z, w has a global connecting path crossing one of finitely many small shells. The minimum of the corresponding internal across distances gives ℓ_k. No connectedness of K is needed.

The hull-arrival function $\tau(z) = \inf\{r : z \in H_r^\circ(0)\}$ is 1-Lipschitz in D: if $\tau(x) < s < t < \tau(y)$, any path from x to y leaves H_s° at a point of root distance s, while $D(0, y) \geq t$, and hence its remaining length is at least $t - s$. The intervals $\tau(Q_k)$ cover $[0, D(0, \partial B_1^{\mathrm{E}})]$. If $0 < v < w < D(0, \partial B_1^{\mathrm{E}})$ and $\operatorname{diam}_D \partial H_t^\circ(0) \leq u$ for almost every $t \in [v, w]$, then

$$w - v \leq \sum_{k\geq 0} U_k \mathbf{1}_{\{\ell_k \leq u\}}. \tag{21}$$

For each interval $\tau(Q_k)$ meeting $[v, w]$ in positive length, choose a radius outside the exceptional null set. The corresponding path crosses ∂H_t°; (20) then forces $\ell_k \leq u$. The remaining zero-length intervals contribute nothing. This argument deliberately avoids treating a point swallowed into a hole as a hull-boundary point.

5. The stationary radial clock

Under the original Gaussian law $\mathbb{P}_0$, write $B_t = h_{e^t}^\circ(0)$ and $W_t = B_t + at$. Define

$$J(t) = \int_{-\infty}^t e^{\xi W_s}\, \mathrm{d}s, \qquad q(t) = \frac{e^{\xi W_t}}{J(t)}, \qquad u(t) = \xi^{-1} \log \frac{J(t)}{J(0)}. \tag{22}$$

Almost surely J is strictly increasing from zero to infinity, so u is a bijection of $\mathbb{R}$ with inverse $t(u)$ and $t(0) = 0$.

Proposition 5.1 (Palm law of the full environment)**.** *The measure $\mathbb{P}_*$ specified by*

$$\frac{\mathrm{d}\mathbb{P}_*}{\mathrm{d}\mathbb{P}_0} = \frac{q(0)}{\xi a} \tag{23}$$

is a probability measure. Under $\mathbb{P}_*$ the field $F = H - \xi^{-1} \log J(0)$ and metric $D_F = J(0)^{-1} D_H$ have a stationary full-environment *evolution in the u clock, given by*

$$F \longmapsto F(e^{t(u)}\cdot) + Qt(u) - u. \tag{24}$$

The independent uniform shell phase is translated by $-t(u)$ modulo one. In particular, the rooted metric law is invariant under a deterministic multiplicative change of metric scale, modulo the indicated Euclidean reembedding.

The process $Y_u = q(t(u))/\xi^2$ is stationary and reversible. Its generator and invariant law are

$$\mathcal{L}_a f(y) = \frac{\xi}{2} y f''(y) + \left(a + \frac{\xi}{2} - \xi y\right) f'(y), \qquad \pi_a(\,\mathrm{d}y) = \frac{2^{1+2a/\xi}}{\Gamma(1+2a/\xi)} y^{2a/\xi} e^{-2y}\,\mathrm{d}y. \tag{25}$$

Furthermore, $t(v) - t(u) = \xi^{-1} \int_u^v Y_w^{-1}\,\mathrm{d}w$.

Proof. The normalized whole-field flow θ_t in Lemma 3.1 is stationary, even though the Brownian coordinate need not be independent of the remaining field. The process $q(t)$ is a stationary functional of this flow and of the Brownian increments prior to t. Applying Itô's formula forward from a deterministic physical time gives

$$\mathrm{d}q = \xi q\,\mathrm{d}B + \left((\xi a + \xi^2/2)q - q^2\right)\,\mathrm{d}t.$$

The unique invariant distribution of q/ξ^2 is Gamma with shape $2a/\xi$ and rate two, by the stationary zero-current equation. Thus $\mathbb{E}_0 q(0) = \xi a$ and (23) is normalized.

The time change has speed $v(t) = \mathrm{d}u/\,\mathrm{d}t = q(t)/\xi$. We spell out the invariant-measure argument for the complete environment. Write θ_t for its physical flow and Ψ_u for the reparameterized flow. For bounded measurable f, set $G_u(\omega) = \int_0^u f(\Psi_w\omega)\,\mathrm{d}w$. The time-change cocycle gives the pathwise identity

$$G_u(\theta_T\omega) - G_u(\omega) = \int_0^T v(\theta_s\omega)\{f(\Psi_u\theta_s\omega) - f(\theta_s\omega)\}\,\mathrm{d}s.$$

The expectation of the left side is zero by invariance under θ_T. By physical stationarity, the expectation of the right side equals $T\{\mathbb{E}_0[v(f\circ\Psi_u)] - \mathbb{E}_0[vf]\}$. Therefore $\mathbb{E}_0[v(f\circ\Psi_u)] = \mathbb{E}_0[vf]$. Thus $v\,\mathrm{d}\mathbb{P}_0/\mathbb{E}_0 v = q(0)\,\mathrm{d}\mathbb{P}_0/(\xi a)$ is invariant under Ψ_u. The argument applies to *every* measurable observable of the complete centered LGF and the translated uniform phase, not merely to functions of q. It proves stationarity of (24). Exact dilation covariance (7) and Weyl scaling identify the metric under this evolution; no independence of radial and nonradial field is used.

Since $\mathrm{d}u/\,\mathrm{d}t = \xi Y$, the diffusion time-change formula gives

$$\mathrm{d}Y_u = \left(a + \frac{\xi}{2} - \xi Y_u\right)\,\mathrm{d}u + \sqrt{\xi Y_u}\,\mathrm{d}\beta_u.$$

The speed bias changes the Gamma shape to $1 + 2a/\xi$, proving (25). The generator is symmetric in $L^2(\pi_a)$ by writing it in divergence form. The inverse-clock identity follows by differentiation. □

Lemma 5.2 (Envelope moments under the Palm law)**.** *Let $M(t) \geq 1$ be a fixed-width local mark of the full normalized physical environment, with $\mathbb{E}_0 M(t)^b < \infty$ uniformly in t, for every $b > 0$. Then under $\mathbb{P}_*$ the stationary convolution*

$$\mathcal{R}(u) = \xi \int_{-\infty}^{u} e^{-\xi(u-v)} M(t(v))\,\mathrm{d}v \tag{26}$$

has moments of every positive order. Taking $M = \max(1, x, y^{-1})$ with (17) yields

$$\frac{S(t(u))}{J(t(u))} \leq \mathcal{R}(u), \qquad \frac{J(t(u))}{A(t(u))} \leq \mathcal{R}(u). \tag{27}$$

In particular $D_F(0, \partial B_1^{\mathrm{E}})$ has negative moments of every order.

Proof. The density in (23) has moments of all positive orders (by its Gamma law). Hölder's inequality transfers the local moments to the Palm law. Stationarity of the complete u-environment then makes them uniform in u. Jensen's inequality for the probability kernel $\xi e^{-\xi(u-v)}\mathbf{1}_{v\leq u}\,\mathrm{d}v$ gives the moment bound on $\mathcal{R}$. Since $e^{\xi W_t}\,\mathrm{d}t = \mathrm{d}J = \xi J\,\mathrm{d}u$, the first inequality in (27) follows by comparing the integrands with M. The second follows from Jensen's inequality for $z \mapsto z^{-1}$. At $t(0) = 0$, the first lower envelope bound in (18) gives $D_F(0, \partial B_1^{\mathrm{E}}) \geq A(0)/J(0)$, and the second ratio in (27) bounds its negative powers. □

Proposition 5.3 (Exact Feynman–Kac exponent)**.** *For $p > 0$ let $\lambda(p) = \sqrt{a^2 + 2p} - a$. The nonnegative-potential Feynman–Kac semigroup satisfies*

$$\left\| \exp\left\{ u(\mathcal{L}_a - p/(\xi y)) \right\} \right\|_{L^2(\pi_a) \to L^2(\pi_a)} = e^{-\lambda(p)u}. \tag{28}$$

If $I_1, \ldots, I_m$ are deterministic, pairwise disjoint intervals in u time, then

$$\mathbb{E}_* \exp\left\{ -p \sum_{j=1}^{m} [t(\sup I_j) - t(\inf I_j)] \right\} \leq \exp\left\{ -\lambda(p) \sum_{j=1}^{m} |I_j| \right\}. \tag{29}$$

Moreover, for $0 < b < a^2/2$,

$$\mathbb{E}_* e^{b(t(U) - t(0))} \leq \exp\{(a - \sqrt{a^2 - 2b})U\}, \qquad U \geq 0. \tag{30}$$

Proof. Set $\nu = (\sqrt{a^2 + 2p} - a)/\xi$ and $\psi(y) = y^\nu$. Direct differentiation shows

$$(\mathcal{L}_a - p/(\xi y))\psi = -\lambda(p)\psi. \tag{31}$$

The reversible Dirichlet form is $\frac{\xi}{2} \int y(f')^2 \, \mathrm{d}\pi_a$. Integration by parts gives, for $f \in C_c^\infty(0, \infty)$,

$$\begin{aligned} \frac{\xi}{2} \int y(f')^2 \, \mathrm{d}\pi_a + \frac{p}{\xi} \int y^{-1} f^2 \, \mathrm{d}\pi_a - \lambda(p) \int f^2 \, \mathrm{d}\pi_a \\ = \frac{\xi}{2} \int y\psi^2 \left((f/\psi)'\right)^2 \mathrm{d}\pi_a \geq 0. \end{aligned} \tag{32}$$

The ground state lies in $L^2(\pi_a)$ and the form domain: near zero its derivative and potential terms have power $2\sqrt{a^2 + 2p}/\xi - 1 > -1$, and the Gamma density controls infinity. Cutoff approximations show it attains the bottom of the spectrum. This proves (28), including its equality; the Feynman–Kac identification follows by monotone truncation of the nonnegative potential.

Order the deterministic intervals I_j. The left side of (29) is an $L^2(\pi_a)$ inner product formed by alternating killed semigroups on the intervals and ordinary reversible Markov semigroups on the gaps. These operators have norms bounded by (28) and one, respectively, and $\|1\|_{L^2(\pi_a)} = 1$. This proves (29) without a conditioning argument.

For (30), repeat the ground-state computation with potential $-b/(\xi y)$ and exponent $(\sqrt{a^2 - 2b} - a)/\xi$. The form-domain integrability holds because $a^2 - 2b > 0$. Truncate the positive potential, apply the resulting lower spectral bound uniformly, and use monotone convergence together with the inverse-clock identity. □

Notice that $Q^2 - a^2 = 2d$. Thus

$$\lambda(d) = \sqrt{a^2 + 2d} - a = Q - a = \gamma, \qquad \frac{\lambda(d)}{\xi} = d_\gamma. \tag{33}$$

This is the sole place where the desired conformal dimension enters the numerical part of the argument.

6. Exceptional scales

The spectral estimate holds for predetermined clock intervals. To use it for geometric annuli, we must show that sufficiently many intervals correspond to good annuli whose Euclidean shapes are controlled. We establish both statements jointly across many scales; individual high-probability estimates would not suffice.

Lemma 6.1 (Simultaneous shape bound)**.** *For any $\varepsilon, \delta, M > 0$, there are $L_0, C < \infty$ such that, for all $L \geq L_0$, $v_0 \in \mathbb{R}$, and $n \geq 1$,*

$$\mathbb{P}_*\left[\#\left\{1 \leq j \leq n : \frac{S(t(v_0+jL))}{J(t(v_0+jL))} > e^{\varepsilon L} \text{ or } \frac{J(t(v_0+jL))}{A(t(v_0+jL))} > e^{\varepsilon L}\right\} \geq \delta n\right] \leq Ce^{-MnL}. \quad (34)$$

Proof. By the full-environment stationarity of Proposition 5.1, take $v_0 = 0$. The ratios are bounded by $\mathcal{R}$ from (26). The contribution to $\mathcal{R}(jL)$ from $v < 0$ is $e^{-\xi jL}\mathcal{R}(0)$. If this exceeds $e^{\varepsilon L}/2$ at at least $\delta n/2$ indices, $\mathcal{R}(0)$ exceeds $c\exp(c_\delta nL)$; arbitrary moments from Lemma 5.2 make this probability $O(e^{-MnL})$ for arbitrary M.

Choose a large C_* so that (30) gives $\mathbb{P}_*[t(nL) > C_*nL] \leq e^{-MnL}$. On the remaining event, only physical unit intervals $[k, k+1]$ with $0 \leq k \leq \lceil C_*nL \rceil$ need consideration. Put

$$H_k = \sup_{t\in[k,k+1]} \log\max(1, x(t), y(t)^{-1}), \quad T_k = H_k + 2\operatorname{osc}_{[k-1,k+1]} W + C_0 + 1.$$

The T_k are fixed-width local Gaussian marks, and $\sup_k \mathbb{E}_0 e^{bT_k} < \infty$ for every $b > 0$. The identity $J(k) \geq \int_{k-1}^k e^{\xi W_s}\,\mathrm{d}s$ gives $u(k+1) - u(k) \leq C_0 + 2\operatorname{osc}_{[k-1,k+1]} W$. If the portion of $\mathcal{R}(v)$ arising from $w \geq 0$ exceeds $e^{\varepsilon L}/2$, there is a witness w with $\log M(t(w)) > \varepsilon L/2 + (\xi/2)(v-w)$. A witness in physical interval $[k, k+1]$ requires $T_k \geq \varepsilon L/2$ and influences at most $C_\varepsilon T_k/L$ grid levels. Hence at least $\delta n/2$ affected levels imply

$$\sum_{0\leq k\leq C_*nL+1} T_k \mathbf{1}_{\{T_k \geq \varepsilon L/2\}} \geq c_\varepsilon \delta nL. \quad (35)$$

For any fixed $b > 0$, Lemma 3.2 and Hölder's inequality for the Palm density give

$$\mathbb{E}_* \exp\left(b\sum_{k=0}^N T_k \mathbf{1}_{\{T_k\geq\varepsilon L/2\}}\right) \leq C\exp\{(N+1)\omega_L(b)\}, \qquad \omega_L(b) \longrightarrow 0.$$

The convergence follows by domination with an exponential moment at a strictly larger parameter. Chernoff's inequality applied to (35), first choosing b large and then L large, gives the specified exponential rate. Combining the initial-history, physical-horizon, and exceptional-mark terms proves the lemma. □

Lemma 6.2 (Separated-level Brownian occupation)**.** *Let $z(t) = \xi(B_t - at)$ for $t \geq 0$, with an arbitrary deterministic initial value. For $0 < c < a/\xi$ and real levels $s_1, \ldots, s_n$ separated pairwise by at least one, put $T_s = \int_0^\infty e^{-c|z(t)-s|}\,\mathrm{d}t$. There are $b, C > 0$, independent of the levels and the initial value, for which*

$$\mathbb{E}\exp\left(b\sum_{j=1}^n T_{s_j}\right) \leq e^{Cn}. \quad (36)$$

Proof. Let $V(x) = \sum_j e^{-c|x-s_j|}$. Separation gives $\|V\|_\infty \leq C_c$, $|V'| \leq cV$ almost everywhere, and $\int_{\mathbb{R}} V = 2n/c$. Let $f(x) = \exp(b_1\int_{-\infty}^x V(y)\,\mathrm{d}y)$. For the generator $\mathcal{A} = (\xi^2/2)\partial_x^2 - a\xi\partial_x$,

$$\frac{\mathcal{A}f}{f} \leq \left(\frac{\xi^2 b_1 c}{2} + \frac{\xi^2 b_1^2\|V\|_\infty}{2} - a\xi b_1\right)V.$$

Choose b_1 small and then $b > 0$ so that $(\mathcal{A} + bV)f \leq 0$. Generalized Itô's formula applies to f, whose derivative is locally absolutely continuous. A positive local supermartingale and Fatou's lemma give (36) using $1 \leq f \leq e^{2b_1 n/c}$. □

Proposition 6.3 (Joint absence of thin tubes)**.** *Fix $0 < A < B < \infty$, $C_0, \zeta > 0$, and $L_f < \infty$. Let f be any possibly field-dependent continuous function with $\|f\|_\infty \leq L_f$, and use D_{F+f}. Put $R_j = R_0\alpha^j$ and $\delta_j = \alpha^\zeta R_j$, $0 \leq j < n$. Let E_j^f be the event*

$$\operatorname{diam}_{D_{F+f}} \partial H_t^\circ(0; D_{F+f}) \leq C_0\delta_j \quad \text{for almost every } t \in [AR_j, BR_j]. \tag{37}$$

For every $\varepsilon, M > 0$, after choosing α sufficiently small,

$$\mathbb{P}_*[\#\{j : E_j^f\} \geq \varepsilon n] \leq C\alpha^{Mn}, \qquad n \geq 1, \tag{38}$$

uniformly in $R_0 > 0$ and in all such perturbations. The same bound holds for a common measurable base-field obstruction which contains the bad-scale events for every admissible f.

Proof. Set $K = \log(1/\alpha)$ and $\Lambda = e^{\xi L_f}$. Weyl comparison gives $\Lambda^{-1}D_F \leq D_{F+f} \leq \Lambda D_F$. Hence the paths and lower barriers in (21) can be estimated for every f by the common base marks ΛU_k and $\Lambda^{-1}\ell_k$. This gives simultaneous obstructions without attempting to compare the filled hull boundaries of two different metrics.

By the full clock-stationarity and exact metric covariance, shift the entire n-scale pattern to $R_j = \alpha^{n+j}$. The bound in Lemma 5.2 implies, for arbitrary M,

$$\mathbb{P}_*[D_F(0, \partial B_1^{\mathrm{E}}) \leq \Lambda B\alpha^n] \leq C_M\alpha^{Mn}.$$

Outside this event the relevant perturbed hulls lie inside B_1^{E}. Thus the inner path used in (21), and no unrestricted outer-shell sum, is sufficient.

Let $r_k = 2^{-k}$, $H = \log 2$, and write, under the physical Gaussian law,

$$m_k = \exp z(kH), \qquad z(t) = \xi(B_{-t} - at), \qquad \widehat{U}_k = U_k/m_k, \quad \widehat{\ell}_k = \ell_k/m_k.$$

Both $\widehat{U}_k$ and $\widehat{\ell}_k^{-1}$ have moments of every positive order, uniformly in $k \geq 0$. In D_F, distances are divided by the original $J(0)$, so the logarithmic threshold levels entering (21) are

$$s_j = c_0 - (n + j + \zeta)K + \log J(0), \qquad 0 \leq j < n,$$

with a deterministic c_0 absorbing C_0 and Λ. Choose $0 < c_1 < \min(1, a/\xi)$ and $c_2 = c_1/2$. The elementary estimate

$$x\mathbf{1}_{\{x\ell \leq 1\}} \leq (1 + \ell^{-1-c_1})e^{-c_1|\log x|} \qquad (x, \ell > 0)$$

then implies that every bad index has a common obstruction

$$\sum_{k \geq 0} W_k e^{-c_1|z(kH) - s_j|} \geq ce^{\zeta K}, \qquad W_k = \widehat{U}_k(1 + \widehat{\ell}_k^{-1-c_1}). \tag{39}$$

The variables W_k are fixed-width Gaussian functionals with all positive moments.

We first bound (39) for deterministic K-separated levels s_j with a common deterministic shift of size at most L_0nK. Put

$$\Omega_k = \sup_{[kH,(k+1)H]} |z(t) - z(kH)|, \qquad Y_k = \log^+ W_k + c_1\Omega_k.$$

They have all exponential moments and satisfy Lemma 3.2. If no k satisfies $Y_k \geq \zeta K/4 + (c_1/2)|z(kH) - s_j|$, the sum in (39) is at most $H^{-1}e^{\zeta K/4}T_{s_j}$ with T as in Lemma 6.2 and $c = c_2$. For large K, the obstruction therefore implies either

$$T_{s_j} > e^{\zeta K/2} \quad \text{or} \quad Y_k \geq \frac{\zeta K}{4} + \frac{c_1}{2}|z(kH) - s_j| \text{ for some } k. \tag{40}$$

The occupation estimate and Markov's inequality show that a positive fraction of the first alternative costs $\exp\{Cn - cne^{\zeta K/2}\}$.

For the second alternative, one index k can influence at most $1 + 4Y_k/(c_1 K)$ of the K-separated levels, and only if $Y_k \geq \zeta K/4$. Thus a positive fraction of bad levels forces

$$\sum_k Y_k \mathbf{1}_{\{Y_k \geq \zeta K/4\}} \geq c_\varepsilon nK.$$

Restrict to $k \leq N = \lceil C_* nK \rceil$. This costs an arbitrarily large exponential rate in nK: Gaussian tails give $z(kH) \leq -a\xi Hk/2$ eventually, and beyond N every relevant level is at distance at least ck from $z(kH)$; an exceptional witness would force $Y_k \geq ck$. For the remaining indices, Lemma 3.2 gives, for fixed $b > 0$ and arbitrarily large $b' > 0$,

$$\log \mathbb{E}_0 \exp\left(b \sum_{k \leq N} Y_k \mathbf{1}_{\{Y_k \geq \zeta K/4\}} \right) \leq C_{b,b'}(N+1) e^{-b'\zeta K/(4p_0)}.$$

Chernoff's inequality, choosing b large and then K large, gives any prescribed $\exp(-MnK)$ bound.

Finally the common shift $\log J(0)$ has exponential tails under $\mathbb{P}_*$, because $q(0) = 1/J(0)$ has a Gamma law with a positive shape parameter. Except on an event of probability Ce^{-MnK}, its magnitude is at most $L_0 nK$. Split this interval into $O(nK)$ unit bins; the kernel inequality changes by at most a fixed multiplicative factor across a bin. Transfer the preceding deterministic-shift estimates from $\mathbb{P}_0$ to $\mathbb{P}_*$ by Hölder's inequality and the positive moments of the density (23). Increasing the Gaussian rates absorbs both the Hölder loss and the $O(nK)$ union bound. The resulting base-field obstruction contains all events (37) for all allowed random f, proving (38). □

7. The rooted product estimate

Fix $p > d$. In the normalized rooted metric D_{F+f}, with f a bounded continuous perturbation, choose $r_j = R_0 \alpha^j$ and $s_j = R_0 \alpha^{j-1+\zeta}$. For any bounded filled hull H containing the origin, define its Euclidean outer and inner radii by

$$\operatorname{outrad}_E H = \sup\{|z| : z \in H\}, \qquad \operatorname{inrad}_E H = \sup\{r : B_r^{\mathrm{E}}(0) \subset H\}.$$

Set

$$v_j^f = \left(\eta + 128 \frac{\operatorname{outrad}_E H^\bullet_{64 r_j}(0; D_{F+f})}{\operatorname{inrad}_E H^\circ_{s_j/2}(0; D_{F+f})} + 2\mathbf{1}_{F_*^f(0,j)^c} \right) \wedge 1. \tag{41}$$

The superscript f in F_*^f means that the event is evaluated using D_{F+f}. All factors are measurable and lie in $[\eta, 1]$.

Proposition 7.1 (Rooted product moment)**.** *Fix $p > d$ and $L_f < \infty$. There are $\zeta \in (0,1)$, $\alpha \in (0,1)$, $\eta \in (0, 1/2)$, and $q > d_\gamma$ such that, for every $R_0 > 0$, $n \geq 1$, and every possibly field-dependent continuous f with $\|f\|_\infty \leq L_f$,*

$$\mathbb{E}_* \left[\left(\prod_{j=1}^n v_j^f \right)^p \right] \leq C\alpha^{qn}. \tag{42}$$

Here the expectation bound is understood with the same parameters for a finite collection of fixed changes to the inner and outer radii, the no-tube threshold, and the positive numerical coefficient multiplying the Euclidean radius ratio. The parameter η may be chosen of the form α^{M_0} with M_0 arbitrarily large.

Proof. Write $K = \log(1/\alpha)$, $\Lambda = e^{\xi L_f}$ and $\beta(p) = \lambda(p)/\xi$. By (33), $\beta(p) > d_\gamma$. Choose small $\delta, \zeta, \varepsilon > 0$ such that

$$\vartheta = (1-\delta)(1-\zeta-4\varepsilon)\beta(p) > d_\gamma, \tag{43}$$

and then $q \in (d_\gamma, \vartheta)$ and M_0 with $M_0 p > \beta(p) + 1$. Finally set $\eta = \alpha^{M_0}$ and take K large.

For each j, introduce the deterministic clock interval

$$I_j = [u_j^-, u_j^+], \qquad u_j^- = \frac{\log(64 r_j) + 2\varepsilon K}{\xi}, \quad u_j^+ = \frac{\log(s_j/2) - 2\varepsilon K}{\xi}. \tag{44}$$

Its length is

$$\ell_K = \frac{(1 - \zeta - 4\varepsilon)K - \log 128}{\xi} > 0.$$

For large K the intervals I_j are pairwise disjoint, and their lower and upper endpoints each form an arithmetic progression with spacing K/ξ.

Call an index j *good* if $F_*^f(0, j)$ occurs and the two envelope ratios in (27) are at most $e^{\varepsilon K}$ at both endpoints u_j^-, u_j^+. Lemma 6.1 and Proposition 6.3 show that, for arbitrary $A_* > 0$, the probability of fewer than $(1 - \delta)n$ good indices is at most $C\alpha^{A_* n}$. The shape estimate is applied to each arithmetic progression with tolerance adjusted by the constant ξ; the no-tube estimate applies with the interval and threshold of (13). Both statements are uniform in deterministic clock translation, R_0, and the allowed f.

At a good index, the lower envelope at $t(u_j^-)$ and the upper envelope at $t(u_j^+)$, together with Weyl comparison and the margins $e^{\varepsilon K} > \Lambda$, give

$$H^{\bullet}_{64 r_j}(0; D_{F+f}) \subset B^{\mathrm{E}}_{e^{t(u_j^-)}}(0), \qquad B^{\mathrm{E}}_{e^{t(u_j^+)}}(0) \subset H^{\circ}_{s_j/2}(0; D_{F+f}).$$

Consequently, with $\Delta_j = t(u_j^+) - t(u_j^-)$,

$$(v_j^f)^p \leq 2^{p-1}(\eta^p + 128^p e^{-p\Delta_j}). \tag{45}$$

For a *deterministic* set G of indices, expand the product of the right-hand side of (45). Proposition 5.3, applied to every deterministic subcollection of the disjoint I_j, yields

$$\mathbb{E}_* \prod_{j \in G} 2^{p-1}(\eta^p + 128^p e^{-p\Delta_j}) \leq \big[2^{p-1}(\eta^p + 128^p e^{-\lambda(p)\ell_K})\big]^{|G|}. \tag{46}$$

There is no conditioning of the radial process on the random set of good indices. Instead, on the event that at least $m = \lceil (1 - \delta)n \rceil$ indices are good, choose any m good indices and discard the other factors, each of which is at most one. Sum (46) over at most 2^n deterministic choices of G. The complement contributes at most $C\alpha^{A_* n}$. We obtain

$$\mathbb{E}_* \left(\prod_{j=1}^{n} v_j^f \right)^p \leq C\alpha^{A_* n} + 2^n \big[2^{p-1}(\alpha^{M_0 p} + 128^p e^{-\lambda(p)\ell_K})\big]^m.$$

The second term is at most $C_1^n \alpha^{\vartheta n}$. Choose $A_* > q$, then increase K so that $C_1 \alpha^{\vartheta - q} \leq 1$. This gives (42). Changing finitely many fixed radius multipliers alters ℓ_K by bounded additive constants, a fixed change in the ratio coefficient alters only the constant C_1, and Proposition 6.3 permits changed fixed thresholds. The strict inequality in (43) absorbs these changes. □

8. Uniform lower bounds for quantum volumes

Let μ_h denote the subcritical γ-GMC measure, normalized by

$$\mu_h(\mathrm{d}x) = \lim_{\varepsilon \downarrow 0} \varepsilon^{\gamma^2/2} e^{\gamma h_\varepsilon(x)} \, \mathrm{d}x,$$

with any fixed smooth regularization convention and a fixed additive normalization. We need uniformity over all centers in a compact set, not merely over a typical point sampled from μ_h.

Proposition 8.1 (Uniform mass of metric balls)**.** *For every compact $K \subset \mathbb{R}^d$ and $\varepsilon > 0$, almost surely there are random $c, r_0 > 0$ such that*

$$\mu_h(B_r^{D_h}(x)) \geq c\, r^{d_\gamma + \varepsilon}, \qquad x \in K, \quad 0 < r < r_0. \tag{47}$$

The assertion persists after adding a continuous field perturbation, with locally adjusted constants.

Proof. Step 1: a metric-to-mass estimate on a grid. Fix a power-of-two integer $M \gg_d 1$. For each dyadic $r \leq 1$ and $z \in (r/M)\mathbb{Z}^d$, set

$$L(z,r) = D_h(\text{across } A_{r/16,r}(z)) + U_{D_h}(A_{r/4,r/3}(z)), \qquad m(z,r) = \mu_h(B^{\mathrm{E}}_{r/64}(z)).$$

The shell moment bounds, after changing from spherical to circle averages, give

$$\sup_{r,z} \mathbb{E}\left[\left(\frac{L(z,r)}{r^{\xi Q} e^{\xi h_r^\circ(z)}}\right)^b\right] < \infty \quad (b > 0).$$

The normalized GMC mass $m(z,r)/(r^{\gamma Q} e^{\gamma h_r^\circ(z)})$ has negative moments of every order; this is the standard subcritical GMC estimate [6]. Since $\gamma Q/d_\gamma = \xi Q$, Hölder's inequality gives

$$\sup_{r,z} \mathbb{E}\left[\left(\frac{L(z,r)}{m(z,r)^{1/d_\gamma}}\right)^b\right] < \infty \quad (b > 0). \tag{48}$$

A compact set contains $O(r^{-d})$ relevant grid points. Markov's inequality and Borel–Cantelli therefore imply, for every $\eta > 0$, on a single probability-one event and all sufficiently small dyadic r,

$$L(z,r) \leq C_\eta r^{-\eta} m(z,r)^{1/d_\gamma} \tag{49}$$

uniformly over these points.

Step 2: uniform upper GMC mass. The GMC structure function is

$$\zeta_\gamma(b) = \left(d + \frac{\gamma^2}{2}\right) b - \frac{\gamma^2 b^2}{2}, \qquad 1 < b < \frac{2d}{\gamma^2}.$$

For every $0 < \beta < \gamma(Q - \sqrt{2d})$, take $b = \sqrt{2d}/\gamma$ (which lies in the moment range). Then $\zeta_\gamma(b) - \beta b > d$. A union bound over an r-grid followed by Borel–Cantelli gives

$$\mu_h(B_r^{\mathrm{E}}(z)) \leq C_\beta r^\beta \tag{50}$$

for every z in a prescribed compact set and all small r. Enlarged grid balls handle arbitrary centers.

Step 3: a diameter estimate for the same Euclidean cube. Let Q_0 be an axis-parallel open cube of side at most one, with closure in a fixed compact region. For any two points $x, y \in Q_0$, construct chains of annular crossings inside Q_0, leading from a common central crossing to x and y. Here are the geometric details needed for uniformity. If a crossing of $A_{r/16,r}(z)$ and a crossing of $A_{r'/16,r'}(z')$ have $r' \in \{r, r/2\}$ and $|z - z'| \leq r/32$, both cross $A_{r/4,r/3}(z)$ and can be connected inside that shell at cost at most twice the second term in $L(z,r)$. Choose geometric interpolation points

$$x_j = c + (1 - 2^{-j})(x - c), \qquad r_j = 2^{-j} r_0, \quad r_0 \asymp \mathrm{side}(Q_0),$$

where c is the center of Q_0 and the implicit constant in r_0 is small depending only on d. Join the successive interpolation points by monotone axis-parallel paths on the (r_j/M)-grid. The number of grid steps per level is bounded, and all their supporting Euclidean r_j-balls lie within Q_0. At a change of scale the grid centers differ by less than $r_j/32$ after increasing M if necessary. The resulting connected crossing chains have tail supports converging to x and y in Euclidean distance, and their metric lengths are summable by the next estimate. They therefore yield paths entirely within Q_0 with common starting crossing and endpoints x, y.

Write $V = \mu_h(Q_0)$. Every mass $m(z, r_j)$ used along these chains is at most V and, by (50), at most $C_\beta r_j^\beta$. Using (49) and a bounded number of marks per scale,

$$\operatorname{diam}_{D_h}(Q_0; Q_0) \leq C \sum_{j \geq 0} r_j^{-\eta} \min\{V, C_\beta r_j^\beta\}^{1/d_\gamma} \leq C' V^{1/d_\gamma - \eta/\beta},$$

provided $0 < \eta < \beta/d_\gamma$. For the last inequality split the geometric sum where $C_\beta r_j^\beta$ crosses V. The event and constants are uniform over all such cubes since the only randomness entered through the fixed countable grid bounds.

Step 4: metric balls. Fix x in a compact set, take r sufficiently small, and let $Q_0 = x + (-s, s)^d$ be the largest centered open cube contained in $B^{D_h}_{r/2}(x)$. Compatibility of the Euclidean and metric topologies implies that its closure stays in the prescribed compact region. Maximality and continuity give some $y \in \partial Q_0$ with $D_h(x, y) = r/2$. Since x, y can be approximated from within Q_0,

$$\frac{r}{2} \leq \operatorname{diam}_{D_h}(Q_0; Q_0) \leq C \mu_h(Q_0)^{1/d_\gamma - \eta/\beta} \leq C \mu_h(B^{D_h}_r(x))^{1/d_\gamma - \eta/\beta}.$$

Choose $\eta = \beta(1/d_\gamma - 1/(d_\gamma + \varepsilon))$. This proves (47) uniformly over x, r. On compact sets, adding a continuous function compares the metric and the mass by bounded Weyl factors, so the statement transfers. □

9. From a rooted estimate to GMC integrals

The rooted estimate is under a speed-biased law for a logarithmically singular field. For the final covering argument we need an almost-sure integral estimate under the ordinary LGF law. This is supplied by the GMC rooting identity, with a localization to control the additive normalization and the random metric scale.

Choose $\rho \in C_c^\infty(\mathbb{R}^d)$ with $\rho \geq 0$ and $\int \rho = 1$, and normalize the LGF by $(h, \rho) = 0$. Its covariance kernel is

$$C(x, z) = -\log|x - z| + g(x) + g(z) + c_0, \qquad g(x) = \int \log|x - y| \rho(y)\, \mathrm{d}y. \tag{51}$$

The function g is smooth with bounded first derivatives: differentiation under the integral is justified in dimension $d \geq 2$ by the local integrability of $|x|^{-1}$, and the gradient is bounded at infinity by the compact support of ρ. The GMC rooting identity [6] is, for nonnegative measurable Φ,

$$\mathbb{E} \int_K \Phi(x, h) \mu_h(\mathrm{d}x) = \int_K w(x) \mathbb{E}\, \Phi(x, h + \gamma C(x, \cdot))\, \mathrm{d}x, \tag{52}$$

where w is positive, smooth, and bounded on compact sets.

For this section take $r_j = \alpha^j$, $s_j = \alpha^{j-1+\zeta}$. Define the Borel, center-dependent factors

$$v_j(x) = \left(\eta + 128 \frac{\operatorname{outrad}_{E,x} H^\bullet_{64 r_j}(x; D_h)}{\operatorname{inrad}_{E,x} H^\circ_{s_j/2}(x; D_h)} + 21 \mathbf{1}_{F_*(x,j)^c} \right) \wedge 1. \tag{53}$$

The subscripts E, x mean Euclidean radii about x, and the event F_* is defined in (13). We use $\eta \leq v_j \leq 1$.

Proposition 9.1 (GMC integral summability)**.** *Fix $p > d$, and choose parameters α, η, ζ in Proposition 7.1, with $q > d_\gamma$. For each $q' \in (d_\gamma, q)$, almost surely, for every compact $K \subset \mathbb{R}^d$ and integer $m \geq 1$,*

$$\sum_{N=m}^{\infty} \alpha^{-q'N} \int_K \left(\prod_{j=m}^{N} v_j(x) \right)^p \mu_h(\mathrm{d}x) < \infty. \tag{54}$$

The same conclusion holds if a finite prefix of the weights is replaced by one.

Proof. Rooted normalization. For a deterministic x, put $h^x = h + \gamma C(x, \cdot)$,

$$g_x(z) = h(x+z) - h_1^\circ(x), \qquad H_x(z) = g_x(z) - \gamma \log|z|,$$

and $c_x = h_1^\circ(x) + \gamma(2g(x) + c_0)$. Translation invariance modulo constants implies that g_x has exactly the circle-pinned LGF law, independent of the deterministic choice of x. Equation (51) gives

$$h^x(x+z) = H_x(z) + c_x + \gamma(g(x+z) - g(x)). \tag{55}$$

Take a smooth cutoff χ equal to one on B_2^{E} and supported in B_3^{E}, and put $f_x(z) = \gamma\chi(z)(g(x+z) - g(x))$. Its supremum norm is bounded by a deterministic L independent of x. In B_2^{E} the rooted field is $H_x + c_x + f_x$.

We explain how the metric at the inserted root is interpreted. On any relatively compact punctured domain, the laws of H_x and of an ordinary LGF plus a bounded smooth function are mutually absolutely continuous, so locality determines the metric there. Its annular positive diameter moments and $a = Q - \gamma > 0$ imply, by Borel–Cantelli, that the diameters of successive inward annuli are summable. The completion therefore adds exactly one point at the insertion. Under the integrated identity (52), a probability-one identity for the original field and metric transfers for Lebesgue-almost every root. By locality on punctured domains and uniqueness of this completion, the metric sampled in (52) agrees with the canonical inserted-root version for the uses below. We do not assert such an identity simultaneously for every deterministic insertion point.

Localization. Define

$$c_h(x) = h_1^\circ(x), \quad J_h(x) = \int_{-\infty}^0 e^{\xi h^\circ_{e^t}(x) + \xi Q t}\,\mathrm{d}t, \quad b_h(x) = \inf_{|z-x|=1} D_h(x,z).$$

All three are continuous in x, and J_h, b_h are positive. By Lemma 3.3, on every compact set there is almost surely an integer M for which

$$\mathcal{L}_M(x;h) = \{M^{-1} \leq J_h(x) \leq M,\ |c_h(x)| \leq M,\ b_h(x) \geq M^{-1}\} \tag{56}$$

holds for all x in that compact set. We initially insert $\mathbf{1}_{\mathcal{L}_M(x;h)}$ in every integral.

Let $J_x = \int_{-\infty}^0 e^{\xi(B_t + at)}\,\mathrm{d}t$ be the exponential integral of g_x, and let J_x^f be the same integral with the additional circle average of f_x. Since $\|f_x\|_\infty \leq L$, we have $e^{-\xi L} J_x \leq J_x^f \leq e^{\xi L} J_x$. The rooted identity (55) gives $J_{h^x}(x) = e^{\xi c_x} J_x^f$ and $c_{h^x}(x) = c_x + (f_x)_1^\circ(0)$. Thus on $\mathcal{L}_M(x; h^x)$,

$$|c_x| \leq M + L, \quad J_x \leq T_M, \qquad \lambda_x := \frac{e^{-\xi c_x}}{J_x} \in [\lambda_M^-, \lambda_M^+] \tag{57}$$

for deterministic $T_M < \infty$ and $0 < \lambda_M^- < \lambda_M^+ < \infty$.

Under the physical Gaussian law of g_x, the Palm law has density $(\xi a J_x)^{-1}$, since its circle average at one is zero. Consequently, for any nonnegative functional Ψ of the canonical rooted field and the uniformly bounded perturbation f_x,

$$\mathbb{E}_0[\mathbf{1}_{\mathcal{L}_M(x;h^x)}\Psi] \leq \xi a T_M\, \mathbb{E}_*\Psi. \tag{58}$$

This does not invoke independence of c_x from g_x.

For fixed M, choose j_M so large that all geometric events at $j \geq j_M$ are confined to $B_1^{\mathrm{E}}(x)$ whenever (56) holds; this follows from the lower bound on $b_{h^x}(x)$ and strict metric-radius margins. Their hulls, diameters, and inner and outer Euclidean radii are therefore determined by the internal metric on $B_2^{\mathrm{E}}(x)$, where (55) holds. Define

$$D_x^f = J_x^{-1} D_{H_x + f_x}.$$

The original h^x-observables at scale r coincide with D_x^f-observables at scale $\lambda_x r$, with λ_x from (57). The rooted product bound applies to D_x^f for every fixed initial scale, uniformly in the bounded (possibly field-dependent) perturbation f_x.

Removing the random scale. Choose $b \in (1, 4/3)$ and divide the deterministic interval $[\lambda_M^-, \lambda_M^+]$ into finitely many multiplicative bins $[\lambda_i, b\lambda_i]$. Within a bin, monotonicity of filled hulls bounds the ratio term in (53) by the corresponding ratio with numerator radius $64b\lambda_i r_j$ and denominator radius $\lambda_i s_j/2$. If F_* fails at the actual multiplier λ_x, then the boundary-diameter inequality holds almost everywhere on the deterministic interval

$$[3b\lambda_i r_{j-1}/16,\ \lambda_i r_{j-1}/4]$$

with threshold $32b\lambda_i s_j$. It is a nonempty subinterval of the actual one precisely because $b < 4/3$. Thus, pathwise, every product of the actual factors is bounded by a sum of finitely many products of fixed-bin majorants. Proposition 7.1, including its fixed-multiplier variants, bounds each of their pth moments by $C\alpha^{q(N-j_M+1)}$. By (58), uniformly over roots x in a compact set,

$$\mathbb{E}\int_K \mathbf{1}_{\mathcal{L}_M(x;h)} \left(\prod_{j=1}^{N} v_j(x)\right)^p \mu_h(\,\mathrm{d}x) \leq C_{K,M}\alpha^{qN}. \tag{59}$$

Here (52) was used, and the finite prefix and number of bins only change the constant.

Multiply (59) by $\alpha^{-q'N}$ and sum in N. Tonelli's theorem gives almost sure summability of the localized integrals for each integer M. On a compact set, some finite random M makes every localization event hold, so the localization can be removed. A countable compact exhaustion and $v_j \geq \eta$ allow arbitrary compact K and deletion of any fixed prefix. This proves (54). □

10. A global quasisymmetric deformation

The deterministic argument in this section is based on the good-annulus and hyperbolic-filling construction of Miller and Tian [2, Sections 5–6]. We give the details needed for the present noncompact and higher-dimensional setting. We now work deterministically with a proper geodesic metric D on $\mathbb{R}^d$, $d \geq 2$, which induces the Euclidean topology. We give a deterministic weighted-filling construction and the geometric admissibility argument in full. Working at every integer scale is important: the target must be one metric on the noncompact space, not a collection of unrelated local metrics.

Fix a sufficiently small $\alpha \in (0,1)$ and put $r_j = \alpha^j$, $s_j = \alpha^{j-1+\zeta}$. Let $\widetilde{F}$ be the intrinsic good-band event defined by (12); it is used only pointwise. Define

$$\sigma(x,j) = \left(\frac{\operatorname{diam}_E B^D_{4r_j}(x)}{\operatorname{inrad}_{E,x} H^\circ_{s_j}(x)} \wedge 1\right)\mathbf{1}_{\widetilde{F}(x,j)} + \mathbf{1}_{\widetilde{F}(x,j)^c}. \tag{60}$$

Lemma 10.1 (Annular admissibility)**.** *Suppose $C_i = B^D_{4r_j}(x_i)$, $0 \leq i \leq N$, are consecutive overlapping metric balls, C_0 meets $B^D_{r_{j-1}}(y)$, and C_N meets $\mathbb{R}^d \setminus B^D_{2r_{j-1}}(y)$. Then*

$$\sum_{i=0}^{N} \sigma(x_i, j) \geq 1. \tag{61}$$

Proof. Write $R = r_{j-1}$, $r = r_j$, $s = s_j$. If any selected weight equals one, the assertion follows. Otherwise every selected center has a witness $\widetilde{F}$, and its weight is the ratio in (60). It is enough to find a consecutive subchain $C_a, \dots, C_b$ such that, with $\mathcal{C} = \bigcup_{i=a}^b C_i$,

$$\mathcal{C} \not\subset H^\circ_s(x_i) \quad \text{for every } i \in [a,b]. \tag{62}$$

Indeed $x_i \in \mathcal{C}$, so $\operatorname{inrad}_{E,x_i} H^\circ_s(x_i) \leq \operatorname{diam}_E \mathcal{C}$; overlapping of the balls implies $\operatorname{diam}_E \mathcal{C} \leq \sum_{i=a}^b \operatorname{diam}_E C_i$. Dividing and summing yields (61).

We describe how to choose the subchain, emphasizing the only topological fact required. Choose $z \in C_N$ with $D(y,z) \geq 2R$. If z lies in the unbounded component of $\mathbb{R}^d \setminus \overline{B^D_{3R/2}(y)}$, work with the nested centered hulls $K^\circ_t = H^\circ_{R+t}(y)$ and $K^\bullet_t = H^\bullet_{R+t}(y)$, and put $a_0 = y$. Otherwise reverse

the chain: let K_t° be the component containing z of $\mathbb{R}^d \setminus \overline{B^D_{2R-t}(y)}$, let $K_t^\bullet$ be the corresponding closed component of $\mathbb{R}^d \setminus B^D_{2R-t}(y)$, and put $a_0 = z$. (Only the closed convention is used at $t = 0$ in the second case.) In either orientation, the sets K_t increase with t, have connected boundaries and connected complements, and the chain crosses from $K_0^\bullet$ to the complement of $K^\circ_{R/2}$. Their boundaries are root-distance levels $R+t$ or $2R-t$. Therefore any path from $K_u^\bullet$ to the complement of K_v° has length at least $v - u$ for $u < v$. The chain can be trimmed at the first exit and last entry across a corridor without changing its overlapping property.

Take the first-exit/last-entry subchain crossing the corridor from $K^\bullet_{2s}$ to the complement of K°_{4s}. Every ball in it meets the closed corridor, so its center is within distance $4r$ of the corridor. Suppose first that $a_0 \notin H^\circ_s(x_i)$ for each center of this subchain. The two sides of the corridor are at distance at least $2s$, and the open s-ball about x_i misses at least one of them. If it misses the outer side, that connected unbounded side is in the unbounded complementary component of the ball. If it misses the inner side, the inner side is connected and contains a_0, which is, by assumption, in that same unbounded complementary component. Thus one endpoint of the selected subchain is outside $H^\circ_s(x_i)$ for every i, proving (62).

Otherwise some selected center x_k swallows a_0, namely $a_0 \in H^\circ_s(x_k)$. The corridor and margin inequalities give

$$K_0^\bullet \subset H^\circ_{R/16}(x_k), \qquad H^\circ_{5R/16}(x_k) \subset K^\circ_{R/2}.$$

For the first inclusion use connectedness of $K_0^\bullet$, its disjointness from $B^D_s(x_k)$, and the fact that it contains a_0; for the second use the connected unbounded exterior of $K^\circ_{R/2}$ and the root-distance margin $R/2 - 4s - 4r > 5R/16$. The witness $\widetilde{F}(x_k, j)$ therefore supplies a *good mixed filled-ball band* $A(o; t, s)$ whose inner hull contains $K_0^\bullet$ and whose outer hull is contained in $K^\circ_{R/2}$. The original chain crosses this band.

Apply the intrinsic arrival coordinate T_A of Lemma 4.2 in this witnessing band. Trim the crossing chain at its first exit from the closed layer $\mathfrak{H}^\bullet_{3s}$ and its last entry into the complement of the open layer $\mathfrak{H}_{5s}$. The two sides are at ambient distance at least $2s$: a path remaining in A pays this amount by the arrival coordinate, and a path leaving A first pays at least $3s$ before hitting I or O. Every retained $4r$-ball meets the middle corridor. Since $4r < s/16$, its center lies in A with T_A between $2s$ and $6s$, by Lemma 4.2. The global s-ball about each such center is consequently an internal s-ball in A, and the good-band condition applies.

Each such s-ball misses at least one of the two sides of the trimmed corridor, since the sides are $2s$ apart. The outer side is connected to the unbounded complement of A without meeting the ball, so if it is missed it lies in the unbounded component of the ball complement. If the inner side is missed, its connectedness and its connection to I within the ball complement, together with goodness, put it in the same unbounded component. In either case the corresponding endpoint of the trimmed chain is outside $H^\circ_s(x_i)$. This proves (62) also in the swallowing case. □

The following comparison allows us to use only the measurable factors $v_j(x)$ in probabilistic arguments. For $k \geq 0$, set

$$\tau_k(x, j) = \left(\eta + 2 \sup_{D(x,y) \leq k r_j} \sigma(y, j) \right) \wedge 1. \tag{63}$$

Lemma 10.2 (Local majorants)**.** *For sufficiently small α,*

$$\tau_{27}(x, j) \leq v_j(x). \tag{64}$$

Moreover, if $D(x, z) \leq r_N/3$ and $j \leq N$, then $\tau_{26}(x, j) \leq \tau_{27}(z, j)$.

Proof. The second inequality follows from the triangle inequality and $r_N \leq r_j$. If $F_*(x, j)$ fails, then $v_j(x) = 1$ and the first inequality is immediate. Otherwise (14) gives $\widetilde{F}(y, j)$ for every

$D(x,y) \leq 27r_j$, provided $27r_j \leq r_{j-1}/16$. Let

$$u = \operatorname{outrad}_{E,x} H^{\bullet}_{64r_j}(x), \qquad v = \operatorname{inrad}_{E,x} H^{\circ}_{s_j/2}(x).$$

By hull inclusion, $B^D_{4r_j}(y) \subset B^D_{31r_j}(x) \subset H^{\bullet}_{64r_j}(x)$, and $H^{\circ}_{s_j/2}(x) \subset H^{\circ}_{s_j}(y)$ if $27r_j \leq s_j/2$. Moreover $|x-y| \leq u$. If $v > 2u$, it follows that $\sigma(y,j) \leq (2u/(v-u)) \wedge 1 \leq 4u/v$. If $v \leq 2u$, the same inequality follows from $\sigma \leq 1$. Hence $\tau_{27}(x,j) \leq (\eta + 8u/v) \wedge 1 \leq v_j(x)$. □

10.1. **A weighted filling at every integer scale.** For each $j \in \mathbb{Z}$, take a maximal r_j-separated set $\mathcal{A}_j \subset (\mathbb{R}^d, D)$ and let $V_j = \{(x,j) : x \in \mathcal{A}_j\}$. The sets are countable and locally finite by properness. Vertices in the same level are neighbors, written $u \sim v$, when their $4r_j$-balls overlap; include self-neighbors. Choose a parent $p(u) \in V_{j-1}$ with center at distance at most r_{j-1} from the center of u. Give the graph its parent edges and its horizontal neighbor edges.

Take $\beta = 32\alpha < 1/2$ and $0 < \eta < \beta$. For $j \leq 0$ put $\pi(u) = \beta^j$. For $j \geq 1$ define inductively

$$\begin{aligned} \nu(u) &= 2\max\{\sigma(w,j) : u \sim v \sim w,\ v,w \in V_j\}, \\ \mu(u) &= \max\{\eta, \nu(u)\} \wedge (1-\eta), \\ \pi_0(u) &= \pi(p(u))\mu(u), \\ \pi(u) &= \max\{\pi_0(u),\ \eta \max_{v\sim u} \pi_0(v)\}. \end{aligned} \tag{65}$$

With $\varrho(u) = \pi(u)/\pi(p(u))$, the finite-neighborhood smoothing calculation gives

$$\eta \leq \varrho(u) \leq 1-\eta\ (j \geq 1), \qquad \eta \leq \frac{\pi(u)}{\pi(v)} \leq \eta^{-1}\ (u \sim v), \qquad \varrho(u) = \beta\ (j \leq 0). \tag{66}$$

To see why it remains valid in this noncompact filling, observe that every maximum in (65) ranges over a finite horizontal neighborhood. Neighboring parents are neighbors one level coarser for small α. The smoothing step raises a preliminary weight only by η times a neighboring preliminary weight and preserves both the child-to-parent bounds and horizontal comparability. The induction is levelwise, and no finiteness of the entire vertex level is used.

The key consequence of admissibility is the horizontal crossing inequality

$$\sum_{i=1}^{N} \min_{w \sim u_{i-1} \text{ or } w \sim u_i} \varrho(w) \geq 1 \tag{67}$$

for any level-j neighbor chain $u_0, \dots, u_N$ crossing a metric annulus of radii $r_{j-1}, 2r_{j-1}$. For $j \geq 1$, if a relevant μ is capped at $1-\eta$, another edge contributes at least η; otherwise the definition of ν and Lemma 10.1 prove the inequality. For $j \leq 0$ all the ratios are β. The chain needs at least $(16\alpha)^{-1}$ edges, since its successive center displacements are at most $8r_j$, hence the sum is at least $(16\alpha)^{-1}\beta \geq 1$.

Give every graph edge (u,v) length $\pi(u)+\pi(v)$, and let d_G be the induced path metric. A fine net approximation to $x \in \mathbb{R}^d$ is a sequence $u_j = (x_j, j)$ with $D(x, x_j) < r_j$. Adjacent approximations at successive levels are connected by a uniformly bounded number of edges and have weights at most $C(1-\eta)^j$, hence they form a d_G-Cauchy sequence. Different choices have vanishing d_G-distance. Write $\iota(x)$ for their common point in the metric completion and define

$$\delta(x,y) = d_G(\iota(x), \iota(y)). \tag{68}$$

Lemma 10.3 (Graph-boundary metric)**.** *The map ι is injective and (68) defines a metric on all of $\mathbb{R}^d$ globally quasisymmetric to D. Moreover, for $N \geq 1$ and $x \in \mathcal{A}_N$,*

$$\operatorname{diam}_\delta B^D_{r_N}(x) \leq C\pi(x,N), \qquad \pi(x,N) \leq (1-\eta)^N. \tag{69}$$

Proof. For vertices u, v, write $g_k(u)$ for their level-k ancestors and let $k(u, v)$ be the greatest level, no larger than either endpoint level, at which $g_k(u)$ and $g_k(v)$ are horizontal neighbors. Such a level exists since ancestor displacements are at most $r_k/(1-\alpha)$ and $r_k \to \infty$ as $k \to -\infty$. Define $P(u, v)$ as the maximum of the weights of their two level-$k(u, v)$ ancestors. Following ancestors down to that level and joining them yields $d_G(u, v) \leq CP(u, v)$, by the geometric decay in (66).

For the reverse inequality, use a finite-path reduction whose only nonlocal input is the horizontal crossing estimate (67). For clarity, give a horizontal edge (u, v) auxiliary length $\min\{\pi_*(u), \pi_*(v)\}$, where $\pi_*(u) = \min_{w \sim u} \pi(w)$, and give a parent edge $(p(u), u)$ auxiliary length $\eta^{-4}\pi_*(u)$. These lengths are at most a fixed multiple of the original edge lengths, by (66). Each excursion above a prescribed level can be replaced by a path at that level without increasing auxiliary length: split its horizontal part at successive first exits from the annuli of its parents. Each crossing pays the corresponding parent weight by (67); the final parent edge pays for any unpaired horizontal edge. Repeat until both endpoints have reached the last level where their ancestors are neighbors. The remaining nontrivial finite path has auxiliary length at least $cP(u, v)$. Thus

$$cP(u, v) \leq d_G(u, v) \leq CP(u, v). \tag{70}$$

Every reduction is performed on a finite path and stops at a finite integer level, so the absence of a lowest level creates no new limit issue.

If $x \neq y$, choose $k \in \mathbb{Z}$ with $r_{k+1} < D(x, y) \leq r_k$. At level k the ancestors of sufficiently fine approximations of x, y are neighbors; at level $k+2$ they are not, provided α is small. Indeed the displacement of an ancestor's center from its limiting point is $O(r_k)$, whereas $D(x, y) > r_{k+1}$ is much larger than r_{k+2}. Hence (70) implies

$$c\pi(z_k(x), k) \leq \delta(x, y) \leq C\pi(z_k(x), k), \tag{71}$$

where $z_k(x)$ is any level-k net point within r_k of x. This proves injectivity. Comparing scales for three distinct points, using (66), gives the quasisymmetric distortion bound

$$\frac{\delta(x, y)}{\delta(x, z)} \leq C \begin{cases} (D(x, y)/D(x, z))^{a_1}, & D(x, y) \leq D(x, z), \\ (D(x, y)/D(x, z))^{a_2}, & D(x, y) > D(x, z), \end{cases} \qquad a_1 = \frac{\log(1-\eta)}{\log \alpha}, \quad a_2 = \frac{\log \eta}{\log \alpha}.$$

Both exponents are positive. Enlarging C produces an increasing homeomorphism of $[0, \infty)$ as distortion function. Finally, if $y, z \in B^D_{r_N}(x)$, their D-distance is at most $2r_N$ and (71), with one neighboring parent step, gives $\delta(y, z) \leq C\pi(x, N)$. The second inequality of (69) is the product of the fine ratios from $\pi(\cdot, 0) = 1$. □

10.2. **Covering sums and proof of the theorem.** The ancestor of a level-N vertex (x, N) at level $j \leq N$ lies within $2r_j$ of x. The two-neighbor maximum in (65) then involves centers at most $26r_j$ from x. From (66), (63), and Lemma 10.2,

$$\pi(x, N) \leq \prod_{j=1}^{N} \tau_{26}(x, j), \qquad \pi(x, N)^p \leq \left(\prod_{j=m}^{N} v_j(z) \right)^p \tag{72}$$

whenever $D(x, z) \leq r_N/3$ and $m \leq N$. The first inequality follows from the smoothing and ancestor comparison, and the second from $\tau_{26} \leq 1$ and (64).

Proof of Theorem 1.1. Fix $p > d$ and the parameters of Proposition 9.1. The deterministic construction above, applied once to $(\mathbb{R}^d, D_h)$, gives one globally quasisymmetric metric δ. Fix compact sets $K \subset \operatorname{int} K' \subset K' \subset \operatorname{int} V$. At level N retain the vertices $A_N \subset \mathcal{A}_N$ whose D_h-balls of radius r_N meet K. Their r_N-balls cover K; for all large N, their centers lie in K'. Their $r_N/3$-balls are disjoint and contained in V.

Choose $d_\gamma + \varepsilon < q' < q$. By the uniform lower mass bound in Proposition 8.1 and the product comparison (72),

$$\sum_{x \in A_N} \pi(x,N)^p \leq C r_N^{-d_\gamma - \varepsilon} \sum_{x \in A_N} \int_{B^{D_h}_{r_N/3}(x)} \left(\prod_{j=m}^{N} v_j(z) \right)^p \mu_h(\,\mathrm{d}z) \tag{73}$$
$$\leq C \alpha^{-(d_\gamma+\varepsilon)N} \int_V \left(\prod_{j=m}^{N} v_j(z) \right)^p \mu_h(\,\mathrm{d}z) \longrightarrow 0.$$

The limit follows from (54), since $q' > d_\gamma + \varepsilon$. By (69), the δ-diameters of the covering balls tend uniformly to zero and their total p-power cost tends to zero. Therefore $\mathcal{H}^p_\delta(K) = 0$. A countable compact exhaustion yields $\mathcal{H}^p_\delta(\mathbb{R}^d) = 0$ for this *same* target metric. We have shown $\mathrm{Cdim}(\mathbb{R}^d, D_h) \leq p$.

Apply the construction on the countable sequence $p = d + 1/k$, $k \geq 1$. The conformal dimension is at most d almost surely. Conversely, any quasisymmetric metric on $\mathbb{R}^d$ induces the Euclidean topology, so its Hausdorff dimension is at least its topological dimension, which equals d. Proposition 2.1 provides a version of the original exponential metric with the exact scaling property used in the radial argument. This establishes (3). □

11. The γ-LQG sphere

Corollary 11.1. *For every $\gamma \in (0,2)$, the ordinary conformal dimension of the intrinsic γ-LQG sphere is two almost surely, for any fixed positive total quantum area.*

Proof. We first work with an unnormalized three-marked sphere. In the field representation of Aru, Huang, and Sun [4], the field before fixing its area is a background GFF on S^2 plus three logarithmic insertions, under a probability law reweighted by a strictly positive finite power of its total GMC mass. On each coordinate disk whose closure avoids the three marked points, this field is locally mutually absolutely continuous with a whole-plane GFF plus a smooth function. The same holds for the corresponding local metric and area measure, by locality, Weyl scaling, and conformal covariance of the γ-LQG metric [5]. Since the reweighting is positive and finite, it does not change probability-one local statements.

Fix $p > 2$. Equip S^2 with an auxiliary smooth metric. The deterministic construction in Section 10 applies to the compact LQG metric, with the finitely many coarse-scale weights set equal to one. At all sufficiently small metric scales, uniformly in the center, a metric ball lies in a smooth coordinate disk of small diameter. We choose a finite coarse-scale cutoff before defining the fine weights; the coarse weights have no effect on the ensuing Hausdorff estimate. Fill the ball toward the unique complementary component containing the exterior of that disk. This convention agrees on smaller coordinate disks with the planar filling convention used in Sections 4 and 10. The good-annulus argument, including its intrinsic arrival coordinate, and the graph-path estimate therefore give one globally quasisymmetric metric δ_p on the entire sphere, with weights $\pi(x,N)$ satisfying the same local product bound as (72). Fixed multiplicative comparisons between smooth and Euclidean distances in a chart are absorbed into the constants in the geometric majorants.

Let K be compact and disjoint from the three insertions. Choose a slightly larger compact set V inside a coordinate disk avoiding the marked points. For all sufficiently fine scales, all the metric balls, filled hulls, and annular observables centered in K are determined by the metric and field in V. The local absolute continuity just described therefore transfers the integral summability statement (54) and the uniform ball-mass estimate of Proposition 8.1 to K. In particular, for

suitable $q' > d_\gamma$ and each sufficiently small $\varepsilon > 0$,

$$\sum_{N\geq 1} \alpha^{-q'N} \int_V \Big(\prod_{j=m}^{N} v_j(z) \Big)^p \mu_h(\,\mathrm{d}z) < \infty, \qquad \mu_h(B_{D_h}(z,r)) \geq c r^{d_\gamma+\varepsilon}$$

for $z \in K$ and all sufficiently small r. Here m may depend on K, to discard the finitely many scales at which the local and global hull conventions differ.

Take the vertices of a global level-N net whose metric r_N-balls meet K. Their $r_N/3$-balls are disjoint and lie in V for large N. The proof of (73), applied to these vertices, gives

$$\sum_x \pi(x,N)^p \leq C\alpha^{-(d_\gamma+\varepsilon)N} \int_V \Big(\prod_{j=m}^{N} v_j(z) \Big)^p \mu_h(\,\mathrm{d}z) \longrightarrow 0.$$

The corresponding covering balls have δ_p-diameters tending uniformly to zero. Hence $\mathcal{H}^p_{\delta_p}(K) = 0$. A countable compact exhaustion of the complement of the marked points, followed by addition of those three points, gives $\mathcal{H}^p_{\delta_p}(S^2) = 0$. Taking $p \downarrow 2$ proves the upper bound; the lower bound follows from the topological dimension of S^2.

Finally, fixing the total quantum area multiplies the LQG metric by one positive random constant. This leaves its conformal dimension unchanged. □

Appendix A. Measurability of the geometric majorants

Only F_* and the ratios v_j are integrated. The existential good-band events F and $\widetilde{F}$, and the pointwise supremum in τ_k, are used in deterministic comparisons and need not be selected measurably. We record the Borel coding for the integrated quantities.

For a proper continuous metric D, compactify $\mathbb{R}^d$ by a point ∞ and set $f_{D,x}(z) = D(x,z)/(1 + D(x,z))$, $f_{D,x}(\infty) = 1$. Properness implies continuity at infinity. The map $(D,x) \mapsto f_{D,x}$ into $C(S^d)$ is Borel since its evaluations on a countable dense set are Borel. The closed superlevel set

$$K_r(D,x) = \left\{ f_{D,x} \geq \frac{r}{1+r} \right\}$$

is a Borel compact-set-valued map. On compact sets containing ∞, taking the component containing ∞ is upper semicontinuous and thus Borel: a Hausdorff limit of connected components containing ∞ is connected, contains ∞, and lies in the corresponding limiting component. The boundary map and the compact-complement operation are Borel in the hyperspace of compact sets; both can be tested by a countable basis of connected open sets. Therefore $H^\circ_r(x)$ and $\partial H^\circ_r(x)$ are Borel. The closed hull is obtained by the decreasing compact-set limit

$$H^\bullet_r(x) = \bigcap_{m\geq 1} H^\circ_{r+1/m}(x).$$

The positive margin in the exterior component of $\{D(x,\cdot) > r\}$ proves this identity. The Euclidean radii of these hulls, and the D-diameters of their compact boundaries, are Borel. The integrand of the Lebesgue integral in (13) is jointly Borel; hence F_* and the factors (53) are jointly Borel in the metric and center. Local versions on relatively compact Euclidean domains follow by truncating the internal distance across a fixed outer collar; all fine hulls agree with these versions once they are contained in a smaller fixed compact set.

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